
Algorithm 1 Meta-evolution of symbolic-regression operators

Input Baseline bundle B_0 ; training tasks $\mathcal{T}$; fitness F ; population target M ; offspring count O ; G generations, including G_s final simplification generations; initial and survivor seed counts $n_0 \leq n_1$.

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1  $\mathcal{P} \leftarrow \{B_0\} \cup \text{INITIALEXPLORE}(B_0, M - 1)$ 
2  $\mathcal{A} \leftarrow \text{EVALUATE}(\mathcal{P}, \mathcal{T}, F, n_0)$  # Store per-task, per-seed scores
3 for  $g = 1, \dots, G$  do
4    $s \leftarrow [g > G - G_s]$  # Switch from performance to simplification
5   if  $s$  and this is the first simplification generation then
6      $\mathcal{P} \leftarrow \text{SEEDFRONTIER}(\mathcal{A}, M)$  # Recover the archive's score-LOC tradeoffs
7   Launch survivor top-ups to  $n_1$  seeds;  $\mathcal{C} \leftarrow \emptyset$ 
8   for  $j$  in a shuffled schedule of  $O$  slots, balanced across operator types do
9      $B \leftarrow \text{TOURNAMENT}(\mathcal{P}, 2)$  # Higher mean fitness wins
10     $m \leftarrow \text{simplify}$  if  $s$ , otherwise a uniform feasible meta-mutation mode
11     $U \leftarrow \text{SOURCEOPERATORS}(\mathcal{P}, B, j, m)$ 
12     $q \leftarrow \text{PROMPT}(j, m, U, \text{OPTIONALTRACEFEEDBACK}(B, \mathcal{T}))$ 
13     $u \leftarrow \text{VALIDATEDLLMCOMPLETION}(q)$  # Sample ensemble; retry invalid code
14     $\mathcal{C} \leftarrow \mathcal{C} \cup \{B[j \leftarrow u]\}$  # Replace one slot; inherit the rest
15  Evaluate  $\mathcal{C}$  on  $n_0$  seeds, overlapping survivor top-ups
16  Merge new observations into  $\mathcal{A}$ ; update task means and overall fitness
17   $\mathcal{P} \leftarrow \begin{cases} \text{SCORELOCSURVIVORS}(\mathcal{P} \cup \mathcal{C}, M), & s, \\ \text{TASKDIVERSESURVIVORS}(\mathcal{P} \cup \mathcal{C}, M), & \text{otherwise} \end{cases}$ 
18  Top up the last survivors to  $n_1$  seeds
19   $\mathcal{Q} \leftarrow \text{FINALISTS}(\mathcal{A}, \mathcal{P})$ ; evaluate  $\mathcal{Q}$  on fresh shared seeds
20 return fresh scores and  $\arg \max_{B \in \mathcal{Q}} \hat{F}_{\text{fresh}}(B)$ 
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Meta-mutation modes are explore, refine, crossover, and simplify. Task-diverse survival retains task champions; score-LOC survival retains nondominated LOC-bucket champions. Both backfill by fitness. Tournament selection uses task-and-seed means; code crossover samples two source operators uniformly. Final identification uses fresh shared seeds on a shortlist of archived bundles.
